IC-307D DAA Lab Project Presentation

Technical Architecture & Algorithm Design Summary

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Slide 1: Executive Curriculum Overview

Course: IC-307D Data Structures & Algorithms Lab | Student: Jayesh Solanki (IC-2K23-45)
Architecture: 7 modular units, 54 automated unit tests, 4 real-world capstone projects, full PEP-8 compliance.

Slide 2: Core Algorithmic Paradigms

- Dynamic Memory Allocation: Pointer-linked lists, dynamic stacks, and unbounded queues.
- Logarithmic Optimizations: AVL tree rotations and min-heap priority queues.
- Graph Theory: Unweighted BFS shortest path and Dijkstra's priority-search algorithm.
- State Restoration: Reversible command pattern with dual undo/redo stacks.

Slide 3: Capstone Architectural Highlights

- Mini Database Engine: Dual indexing with hash primary lookup O(1) and BST secondary range scans O(log n).
- Clinical Hospital System: Priority queue triage prioritizing emergency codes, paired with epidemiological contact tracing.
- Pathfinding Visualizer: ASCII grid representation comparing BFS expansion and Dijkstra shortest distance.

Slide 4: Key Learning Outcomes

Deep mathematical intuition for time/space asymptotic complexity, zero memory-leak dynamic pointer programming, and test-driven development (TDD) principles.